

Assignment 2 – Task 2: Estimation and Interpretation of an ERGM

Glasgow Wave 1 Friendship Network

2026-05-02

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Data Preparation

```
f1 <- as.matrix(read.csv(file.path(DATA, "f1.csv"), header = FALSE))

demo    <- read.csv(file.path(DATA, "demographic.csv"), header = TRUE)
alc_raw <- read.csv(file.path(DATA, "alcohol.csv"),      header = TRUE)
dist_mat <- as.matrix(read.csv(file.path(DATA, "logdistance.csv"), header = FALSE))

n <- nrow(f1)

demo$boy    <- ifelse(demo$gender == 1, 1, 0)
demo$alcohol <- alc_raw$Wave.1

net <- network(f1, directed = TRUE)
network::set.vertex.attribute(net, "gender", demo$gender)
network::set.vertex.attribute(net, "boy",    demo$boy)
network::set.vertex.attribute(net, "alcohol", demo$alcohol)
network::set.vertex.attribute(net, "age",    demo$age)
net

## Network attributes:
##   vertices = 129
##   directed = TRUE
##   hyper = FALSE
##   loops = FALSE
##   multiple = FALSE
##   bipartite = FALSE
##   total edges= 448
##   missing edges= 0
##   non-missing edges= 448
##
## Vertex attribute names:
##   age alcohol boy gender vertex.names
##
## No edge attributes

outdeg <- rowSums(f1)
indeg  <- colSums(f1)

desc <- data.frame(
  Statistic = c("Nodes", "Directed ties", "Density",
               "Mean out-degree", "SD out-degree",
               "Mean in-degree", "SD in-degree"),
  Value = c(n, sum(f1), round(sum(f1) / (n*(n-1)), 4),
           round(mean(outdeg), 3), round(sd(outdeg), 3),
           round(mean(indeg), 3), round(sd(indeg), 3))
)
```

```
kable(desc, caption = "Basic descriptives of the Glasgow Wave 1 network.")
```

Basic descriptives of the Glasgow Wave 1 network.

Statistic	Value
Nodes	129.0000
Directed ties	448.0000
Density	0.0271
Mean out-degree	3.4730
SD out-degree	1.6300
Mean in-degree	3.4730
SD in-degree	2.2680

Part (1): ERGM with Edges and Gender Homophily

Model Specification and Estimation

```
M1 <- ergm(
  net ~ edges + nodematch("gender"),
  control = control.ergm(seed = 42)
)
summary(M1)

## Call:
## ergm(formula = net ~ edges + nodematch("gender"), control = control.ergm(
## seed = 42))
##
## Maximum Likelihood Results:
##
##              Estimate Std. Error MCMC % z value Pr(>|z|)
## edges          -5.0133    0.1365      0  -36.72  <1e-04 ***
## nodematch.gender  2.0098    0.1460      0   13.77  <1e-04 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
##      Null Deviance: 22890 on 16512 degrees of freedom
## Residual Deviance:  3824 on 16510 degrees of freedom
##
## AIC: 3828 BIC: 3843 (Smaller is better. MC Std. Err. = 0)
```

Conditional Probability of a Same-Gender Tie

In an ERGM the conditional probability that actor i nominates actor j , given the rest of the network, is

$$P(x_{ij} = 1 | x_{-ij}) = \text{logistic}(\boldsymbol{\theta}^T \boldsymbol{\delta}(x, i \rightarrow j))$$

For this two-parameter model the change statistic for adding tie $i \rightarrow j$ is:

$$\boldsymbol{\delta} = \left(1, \mathbf{1}[\text{gender}_i = \text{gender}_j]\right)$$

so the conditional probabilities simplify to:

$$P(x_{ij} = 1 | \text{same gender}) = \text{logistic}(\hat{\theta}_{\text{edges}} + \hat{\theta}_{\text{gender}})$$

$$P(x_{ij} = 1 | \text{different gender}) = \text{logistic}(\hat{\theta}_{\text{edges}})$$

```
th_e <- coef(M1)["edges"]
th_g <- coef(M1)["nodematch.gender"]

p_same <- plogis(th_e + th_g)
p_diff <- plogis(th_e)
```

```

cat(sprintf("edges coefficient      : %+.4f\n", th_e))
## edges coefficient      : -5.0133
cat(sprintf("nodematch(gender) coeff : %+.4f\n", th_g))
## nodematch(gender) coeff : +2.0098
cat(sprintf("P(tie | same gender)    : %.5f\n", p_same))
## P(tie | same gender)    : 0.04726
cat(sprintf("P(tie | diff gender)    : %.5f\n", p_diff))
## P(tie | diff gender)    : 0.00660
cat(sprintf("Odds ratio (same/diff)   : %.3f\n", exp(th_g)))
## Odds ratio (same/diff)   : 7.462

```

Interpretation. The estimated edges coefficient is -5.013, reflecting the very low baseline density of the network (around 2.7%). The nodematch("gender") coefficient is 2.01 (exp = 7.462), indicating that a friendship nomination between two students of the same gender is more likely than between students of different genders. Concretely, the model-implied conditional probability of a tie between two same-gender students is $\hat{p}_{\text{same}} = 0.04726$, compared with $\hat{p}_{\text{diff}} = 0.0066$ for mixed-gender dyads – a ratio of approximately 7.16:1. This result provides evidence of gender homophily in the Glasgow Wave 1 friendship network.

Part (2): Extended ERGM Specification

We augment M1 with effects that simultaneously test the seven hypotheses listed in the assignment.

Mapping of hypotheses to ERGM terms in M2.

Hypothesis	ERGM term	Expected sign
i. Reciprocity	mutual	+
ii. Transitivity	gwesp(0.6, fixed=TRUE)	+
iii. Boys higher activity	nodecov('boy')	+
iv. Boys higher popularity	nodeicov('boy')	+
v. High-alcohol higher activity	nodecov('alcohol')	+
vi. High-alcohol higher popularity	nodeicov('alcohol')	+
vii. Alcohol homophily	absdiff('alcohol')	-

Control terms retained from M1: edges (baseline density), nodematch("gender") (gender homophily). The decay parameter for gwesp is fixed at $\alpha = 0.3$.

Model formula:

```
M2_formula <- net ~ edges +
  mutual +
  nodematch("gender") +
  nodecov("boy") +
  nodeicov("boy") +
  nodecov("alcohol") +
  nodeicov("alcohol") +
  absdiff("alcohol") +
  edgecov(dist_mat) +
  gwesp(0.6, fixed = TRUE)
```

Part (3): Estimation and Convergence

Estimation

```
M2 <- ergm(M2_formula,
  control = control.ergm(
    seed = 42,
    MCMC.burnin = 2^16,
    MCMC.interval = 2^10,
    MCMC.samplesize = 5000,
    MCMLE.maxit = 6,
    parallel = 4
  ))

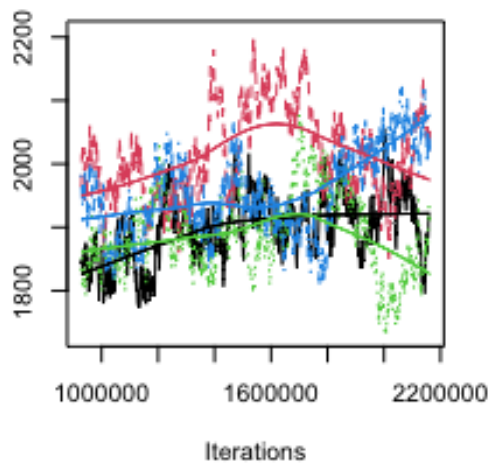
summary(M2)

## Call:
## ergm(formula = M2_formula, control = control.ergm(seed = 42,
##      MCMC.burnin = 2^16, MCMC.interval = 2^10, MCMC.samplesize = 5000,
##      MCMLE.maxit = 6, parallel = 4))
##
## Monte Carlo Maximum Likelihood Results:
##
##              Estimate Std. Error MCMC % z value Pr(>|z|)
## edges          -6.66538    0.20873    0 -31.934 < 1e-04 ***
## mutual           2.96486    0.10380    0  28.562 < 1e-04 ***
## nodematch.gender  0.67331    0.05439    0  12.380 < 1e-04 ***
## nodecov.boy       -0.30307    0.09169    0  -3.305 0.000949 ***
## nodecov.boy        0.12679    0.09160    0   1.384 0.166335
## nodecov.alcohol   -0.17473    0.02893    0  -6.040 < 1e-04 ***
## nodecov.alcohol    0.20868    0.03376    0   6.181 < 1e-04 ***
## absdiff.alcohol   -0.14778    0.03600    0  -4.106 < 1e-04 ***
## edgecov.dist_mat  -0.05186    0.03342    0  -1.552 0.120653
## gwesp.OTP.fixed.0.6 1.84546    0.09913    0  18.616 < 1e-04 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
##      Null Deviance: 22890  on 16512  degrees of freedom
## Residual Deviance:  3470  on 16502  degrees of freedom
##
## AIC: 3490  BIC: 3567  (Smaller is better. MC Std. Err. = 1.995)
```

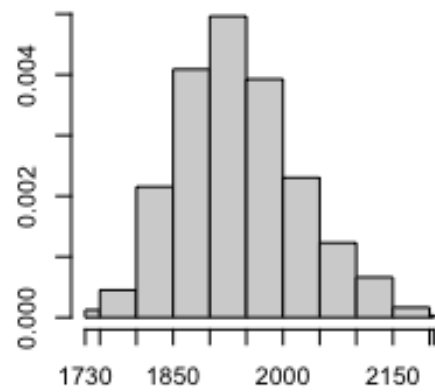
Convergence Diagnostics

```
mcmc.diagnostics(M2)
```

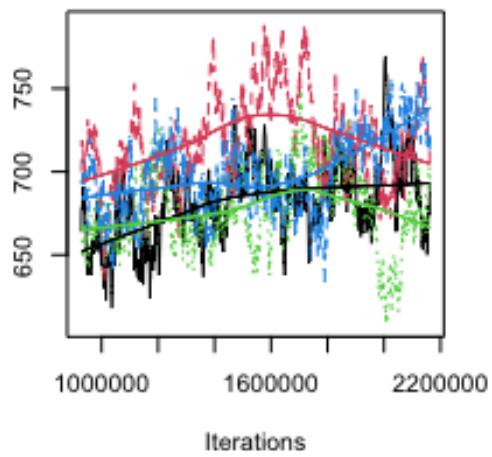

Trace of edges



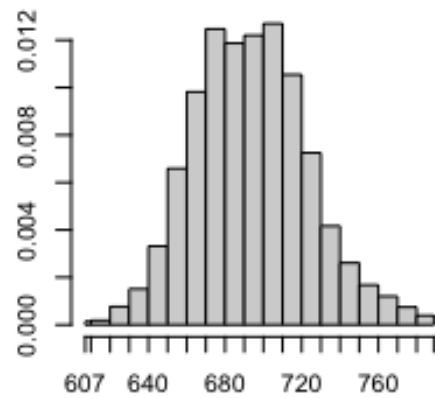
Density of edges



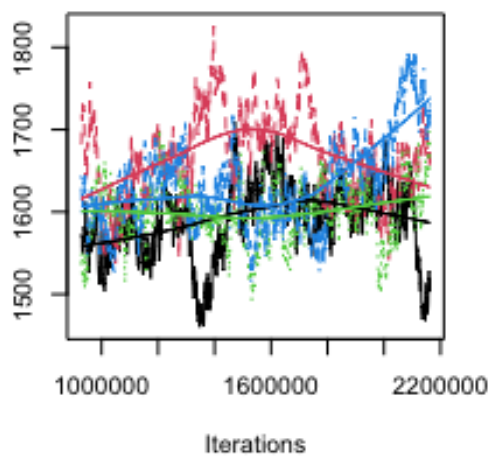
Trace of mutual



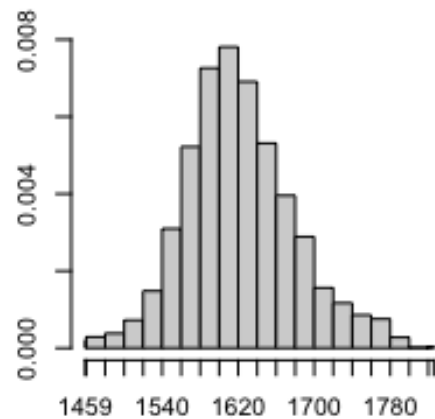
Density of mutual



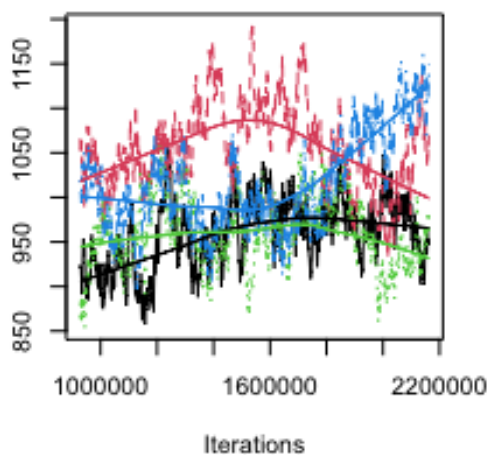
Trace of nodematch.gender



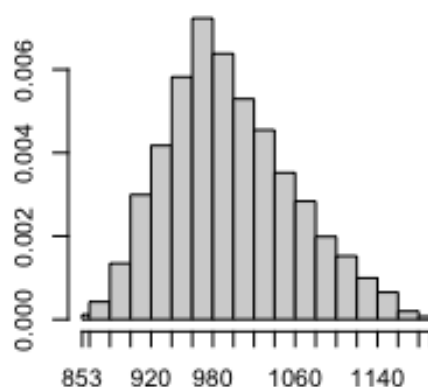
Density of nodematch.gender



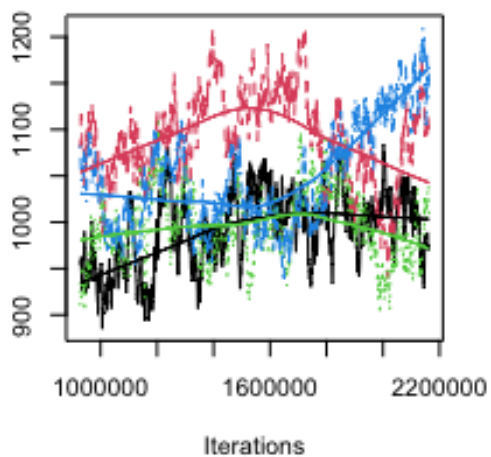
Trace of nodecov.boy



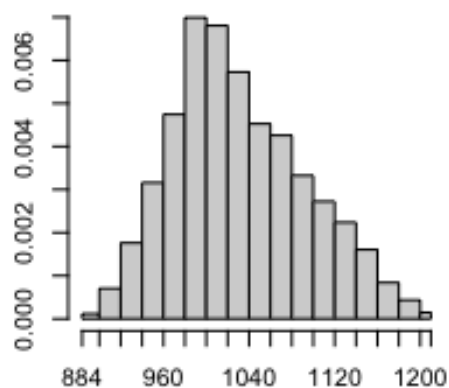
Density of nodecov.boy



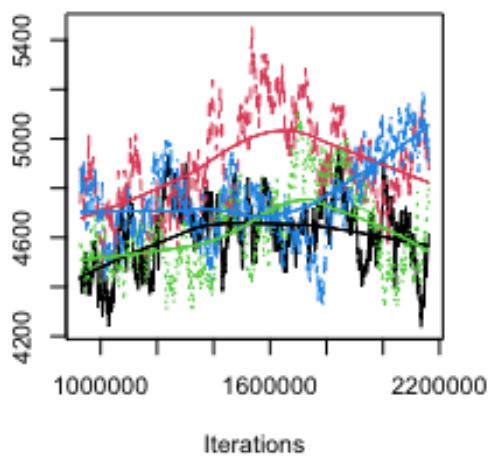
Trace of nodeicov.boy



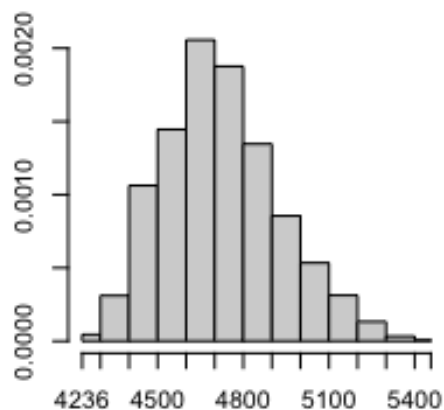
Density of nodeicov.boy



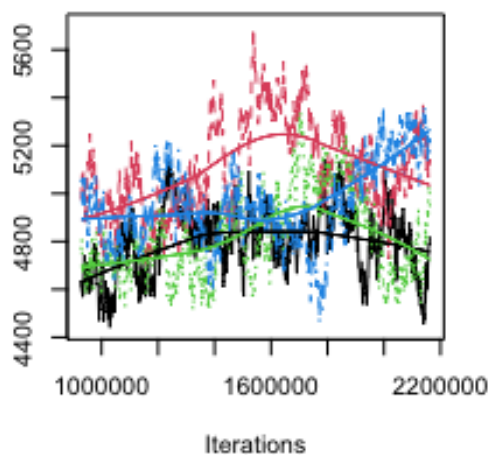
Trace of nodecov.alcohol



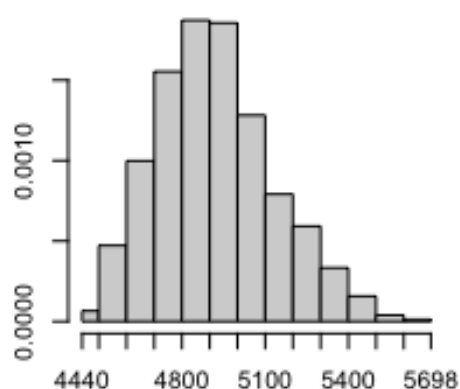
Density of nodecov.alcohol



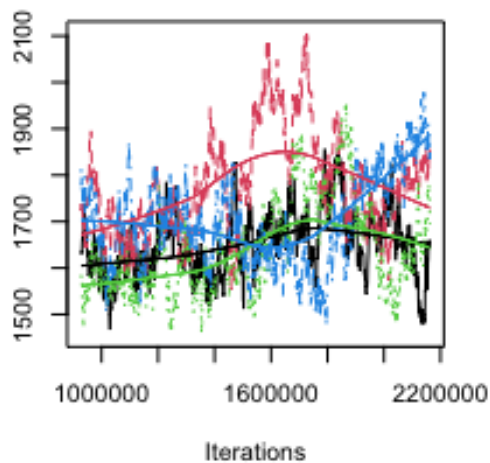
Trace of nodeicov.alcohol



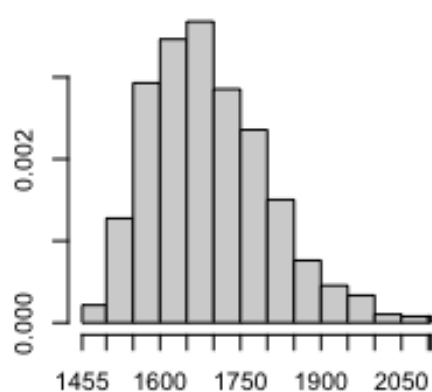
Density of nodeicov.alcohol



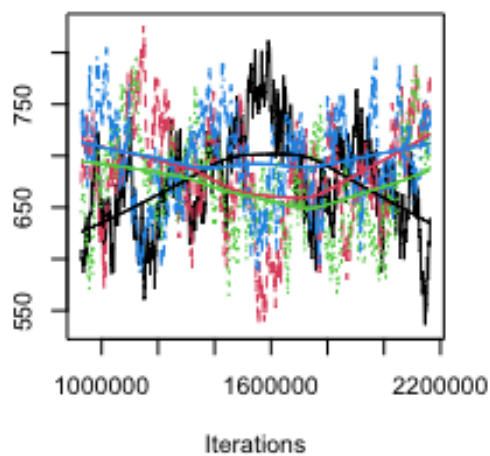
Trace of absdiff.alcohol



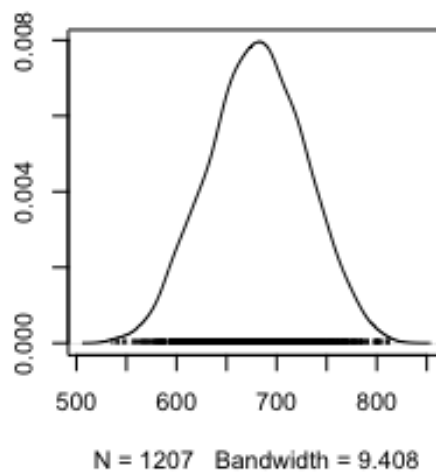
Density of absdiff.alcohol



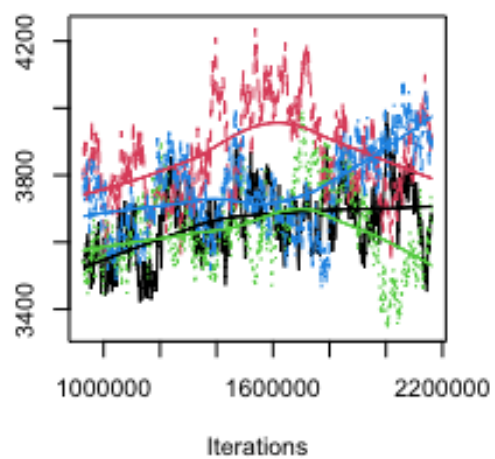
Trace of edgecov.dist_mat



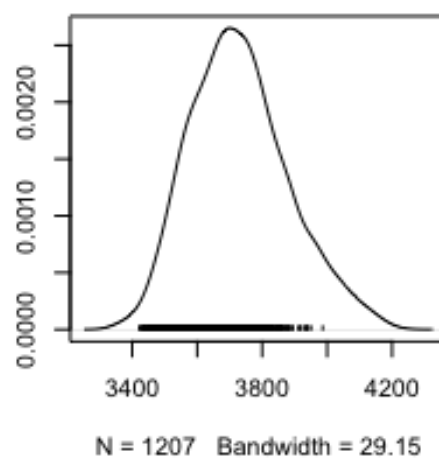
Density of edgecov.dist_mat



Trace of gwesp.OTP.fixed.0.6



Density of gwesp.OTP.fixed.0.6



```

## Sample statistics summary:
##
## Iterations = 927744:2162688
## Thinning interval = 1024
## Number of chains = 4
## Sample size per chain = 1207
##
## 1. Empirical mean and standard deviation for each variable,
##    plus standard error of the mean:
##
##              Mean      SD Naive SE Time-series SE
## edges          1939.4  81.54   1.1735      9.338
## mutual          694.2  29.69   0.4273      2.546
## nodematch.gender 1622.4  57.39   0.8259      5.791
## nodecov.boy       997.2  61.46   0.8845      5.645
## nodeicov.boy     1031.8  61.96   0.8917      6.625
## nodecov.alcohol  4723.2 201.81   2.9044     25.075
## nodeicov.alcohol 4920.5 212.36   3.0563     22.503
## absdiff.alcohol  1692.8 110.76   1.5940     12.774
## edgescov.dist_mat  680.3  48.41   0.6968      5.166
## gwesp.OTP.fixed.0.6 3730.1 151.56   2.1812     16.438
##
## 2. Quantiles for each variable:
##
##              2.5%    25%    50%    75%   97.5%
## edges          1800.7 1881.0 1932.0 1989.0 2117.3
## mutual          640.7  673.0  693.0  714.0  760.0
## nodematch.gender 1518.0 1584.0 1618.0 1657.0 1757.0
## nodecov.boy       893.0  953.0  989.0 1037.0 1130.3
## nodeicov.boy       928.0  987.0 1022.5 1073.2 1163.0
## nodecov.alcohol  4382.7 4580.0 4705.0 4848.0 5166.3
## nodeicov.alcohol 4568.0 4769.0 4904.0 5051.0 5386.6
## absdiff.alcohol  1519.0 1610.0 1680.0 1764.0 1955.0
## edgescov.dist_mat  587.0  647.5  680.6  714.3  773.7
## gwesp.OTP.fixed.0.6 3471.7 3621.1 3718.6 3822.1 4062.8
##
##
## Are sample statistics significantly different from observed?
##              edges    mutual nodematch.gender nodecov.boy nodeicov.boy
## diff.          1939.4174 694.2123      1622.3724      997.2028      1031.8045
## test stat.    185.3341 234.5518      284.3007      151.1501      152.5247
## P-val.         0.0000   0.0000      0.0000      0.0000      0.0000
##
##              nodecov.alcohol nodeicov.alcohol absdiff.alcohol edgescov.dist_
mat
## diff.          4723.1518      4920.5383      1692.7821      680.2
766
## test stat.      181.7474      193.4819      126.1982      129.7
980
## P-val.          0.0000      0.0000      0.0000      0.0
000

```

```

##          gwesp.OTP.fixed.0.6          (Omni)
## diff.          3730.0801          NA
## test stat.          203.7424  3.535974e+05
## P-val.          0.0000  7.618451e-232
##
## Sample statistics cross-correlations:
##          edges      mutual  nodematch.gender  nodeocov.boy
## edges          1.0000000  0.9142453          0.8050760  0.9039778
## mutual          0.9142453  1.0000000          0.7718956  0.8271251
## nodematch.gender  0.8050760  0.7718956          1.0000000  0.8475571
## nodeocov.boy      0.9039778  0.8271251          0.8475571  1.0000000
## nodeicov.boy      0.9084224  0.8318157          0.8511429  0.9647821
## nodeocov.alcohol  0.9359397  0.8592913          0.7453043  0.8196147
## nodeicov.alcohol  0.9416727  0.8478970          0.7459562  0.8335683
## absdiff.alcohol   0.8349956  0.7835165          0.7445729  0.7557004
## edg cov.dist_mat  0.1453105  0.1366065          0.2642176  0.1890089
## gwesp.OTP.fixed.0.6 0.9970111  0.9174596          0.8104210  0.8966124
##          nodeicov.boy  nodeocov.alcohol  nodeicov.alcohol
## edges          0.9084224          0.93593969          0.94167272
## mutual          0.8318157          0.85929132          0.84789696
## nodematch.gender  0.8511429          0.74530428          0.74595617
## nodeocov.boy      0.9647821          0.81961472          0.83356831
## nodeicov.boy      1.0000000          0.82869171          0.83382047
## nodeocov.alcohol  0.8286917          1.00000000          0.98345441
## nodeicov.alcohol  0.8338205          0.98345441          1.00000000
## absdiff.alcohol   0.7563700          0.86824735          0.87506017
## edg cov.dist_mat  0.1946437          0.05878472          0.05755798
## gwesp.OTP.fixed.0.6 0.9024061          0.93546212          0.93936147
##          absdiff.alcohol  edg cov.dist_mat  gwesp.OTP.fixed.0.6
## edges          0.8349956          0.14531054          0.9970111
## mutual          0.7835165          0.13660646          0.9174596
## nodematch.gender  0.7445729          0.26421758          0.8104210
## nodeocov.boy      0.7557004          0.18900889          0.8966124
## nodeicov.boy      0.7563700          0.19464369          0.9024061
## nodeocov.alcohol  0.8682474          0.05878472          0.9354621
## nodeicov.alcohol  0.8750602          0.05755798          0.9393615
## absdiff.alcohol   1.0000000          0.03710850          0.8322541
## edg cov.dist_mat  0.0371085          1.00000000          0.1464004
## gwesp.OTP.fixed.0.6 0.8322541          0.14640041          1.0000000
##
## Sample statistics auto-correlation:
## Chain 1
##          edges      mutual  nodematch.gender  nodeocov.boy  nodeicov.boy
## Lag 0      1.0000000  1.0000000          1.0000000  1.0000000  1.0000000
## Lag 1024  0.9518628  0.9549608          0.9523490  0.9512996  0.9493788
## Lag 2048  0.9099949  0.9145685          0.9113027  0.9090131  0.9084567
## Lag 3072  0.8727191  0.8777489          0.8760645  0.8680689  0.8701725
## Lag 4096  0.8378783  0.8456267          0.8459807  0.8311686  0.8318723
## Lag 5120  0.8079858  0.8134788          0.8216820  0.8017050  0.8015772
##          nodeocov.alcohol  nodeicov.alcohol  absdiff.alcohol  edg cov.dist_mat

```

```

t
## Lag 0      1.0000000      1.0000000      1.0000000      1.0000000
0
## Lag 1024   0.9535639      0.9476797      0.9502314      0.961668
8
## Lag 2048   0.9124189      0.9028816      0.9037955      0.928976
0
## Lag 3072   0.8757569      0.8647426      0.8665877      0.901081
1
## Lag 4096   0.8435612      0.8295993      0.8343233      0.872118
4
## Lag 5120   0.8156835      0.7986835      0.8039214      0.846583
2
##          gwesp.OTP.fixed.0.6
## Lag 0      1.0000000
## Lag 1024   0.9422326
## Lag 2048   0.8935127
## Lag 3072   0.8520133
## Lag 4096   0.8137904
## Lag 5120   0.7811149
## Chain 2
##          edges      mutual nodematch.gender nodecov.boy nodeicov.boy
## Lag 0      1.0000000 1.0000000      1.0000000      1.0000000      1.0000000
## Lag 1024  0.9716893 0.9622623      0.9630436      0.9655540      0.9643913
## Lag 2048  0.9473884 0.9310448      0.9309673      0.9353069      0.9332730
## Lag 3072  0.9245907 0.9052395      0.8999722      0.9065439      0.9055164
## Lag 4096  0.9029565 0.8813885      0.8715949      0.8789031      0.8831274
## Lag 5120  0.8818828 0.8590059      0.8450890      0.8559570      0.8615053
##          nodecov.alcohol nodeicov.alcohol absdiff.alcohol edgescov.dist_ma
t
## Lag 0      1.0000000      1.0000000      1.0000000      1.0000000
0
## Lag 1024   0.9722920      0.9707240      0.9774292      0.964810
2
## Lag 2048   0.9481430      0.9458564      0.9586081      0.932038
1
## Lag 3072   0.9254213      0.9226394      0.9398224      0.901722
9
## Lag 4096   0.9038695      0.9011128      0.9224798      0.873163
4
## Lag 5120   0.8836864      0.8818994      0.9055160      0.846213
0
##          gwesp.OTP.fixed.0.6
## Lag 0      1.0000000
## Lag 1024   0.9652289
## Lag 2048   0.9363728
## Lag 3072   0.9101936
## Lag 4096   0.8858046
## Lag 5120   0.8632507
## Chain 3

```

```

##          edges      mutual nodematch.gender nodecov.boy nodeicov.boy
## Lag 0      1.0000000 1.0000000      1.0000000      1.0000000      1.0000000
## Lag 1024 0.9646810 0.9560466      0.9212413      0.9521877      0.9457040
## Lag 2048 0.9333214 0.9179030      0.8537861      0.9073240      0.8968157
## Lag 3072 0.9039075 0.8828601      0.7935029      0.8688609      0.8507035
## Lag 4096 0.8810653 0.8516527      0.7438427      0.8360683      0.8130038
## Lag 5120 0.8599247 0.8195074      0.6935156      0.8051884      0.7784307
##          nodecov.alcohol nodeicov.alcohol absdiff.alcohol edgescov.dist_ma
t
## Lag 0      1.0000000      1.0000000      1.0000000      1.0000000
0
## Lag 1024      0.9670046      0.9658444      0.9747431      0.959059
9
## Lag 2048      0.9358705      0.9367329      0.9522350      0.925854
8
## Lag 3072      0.9064626      0.9096851      0.9300275      0.895561
1
## Lag 4096      0.8827375      0.8881471      0.9118460      0.867542
1
## Lag 5120      0.8621597      0.8667166      0.8930388      0.841288
1
##          gwesp.OTP.fixed.0.6
## Lag 0      1.0000000
## Lag 1024      0.9555721
## Lag 2048      0.9177470
## Lag 3072      0.8829641
## Lag 4096      0.8573087
## Lag 5120      0.8331580
## Chain 4
##          edges      mutual nodematch.gender nodecov.boy nodeicov.boy
## Lag 0      1.0000000 1.0000000      1.0000000      1.0000000      1.0000000
## Lag 1024 0.9685617 0.9508594      0.9639555      0.9695019      0.9738637
## Lag 2048 0.9409953 0.9048198      0.9333851      0.9426917      0.9519020
## Lag 3072 0.9171577 0.8621545      0.9078853      0.9192761      0.9336786
## Lag 4096 0.8945098 0.8248870      0.8827620      0.8958448      0.9173464
## Lag 5120 0.8757830 0.7897799      0.8628310      0.8765462      0.9026859
##          nodecov.alcohol nodeicov.alcohol absdiff.alcohol edgescov.dist_ma
t
## Lag 0      1.0000000      1.0000000      1.0000000      1.0000000
0
## Lag 1024      0.9653671      0.9645277      0.9709168      0.954605
1
## Lag 2048      0.9344065      0.9343893      0.9449216      0.915643
8
## Lag 3072      0.9075433      0.9086638      0.9206225      0.882082
7
## Lag 4096      0.8841009      0.8847260      0.8988385      0.851621
9
## Lag 5120      0.8635773      0.8643403      0.8798100      0.818832
5

```



```

##          gwesp.OTP.fixed.0.6
## Lag 0          1.0000000
## Lag 1024       0.9593126
## Lag 2048       0.9239773
## Lag 3072       0.8949904
## Lag 4096       0.8677077
## Lag 5120       0.8460456
##
## Sample statistics burn-in diagnostic (Geweke):
## Chain 1
##
## Fraction in 1st window = 0.1
## Fraction in 2nd window = 0.5
##
##          edges          mutual    nodematch.gender    nodeeco
v.boy
##          -7.562091          -5.041203          -4.343847          -8.0
67808
##          nodeicov.boy    nodeocov.alcohol    nodeicov.alcohol    absdiff.al
cohol
##          -8.255321          -5.814197          -4.852017          -2.1
65307
##          edgecov.dist_mat gwesp.OTP.fixed.0.6
##          -1.650082          -7.838045
##
## Individual P-values (lower = worse):
##          edges          mutual    nodematch.gender    nodeeco
v.boy
##          3.966399e-14          4.626138e-07          1.400092e-05          7.15712
1e-16
##          nodeicov.boy    nodeocov.alcohol    nodeicov.alcohol    absdiff.al
cohol
##          1.514942e-16          6.092574e-09          1.222124e-06          3.03641
9e-02
##          edgecov.dist_mat gwesp.OTP.fixed.0.6
##          9.892615e-02          4.576168e-15
## Joint P-value (lower = worse): 0.003859698
## Chain 2
##
## Fraction in 1st window = 0.1
## Fraction in 2nd window = 0.5
##
##          edges          mutual    nodematch.gender    nodeeco
v.boy
##          -1.9227472          -1.7055693          -0.7065637          -1.29
71462
##          nodeicov.boy    nodeocov.alcohol    nodeicov.alcohol    absdiff.al
cohol
##          -0.8762352          -3.2949443          -2.9858985          -1.85
28955

```

```

##      edg cov.dist_mat gwesp.OTP.fixed.0.6
##      0.2661378      -2.0579938
##
## Individual P-values (lower = worse):
##      edges      mutual      nodematch.gender      nodeeco
v.boy
##      0.0545118071      0.0880882966      0.4798376352      0.19458
08775
##      nodeicov.boy      nodeocov.alcohol      nodeicov.alcohol      absdiff.al
cohol
##      0.3809021765      0.0009844119      0.0028274651      0.06389
73367
##      edg cov.dist_mat gwesp.OTP.fixed.0.6
##      0.7901330883      0.0395907293
## Joint P-value (lower = worse): 0.006428974
## Chain 3
##
## Fraction in 1st window = 0.1
## Fraction in 2nd window = 0.5
##
##      edges      mutual      nodematch.gender      nodeeco
v.boy
##      -1.14911064      -2.27697751      -0.30508742      -1.337
32088
##      nodeicov.boy      nodeocov.alcohol      nodeicov.alcohol      absdiff.al
cohol
##      -1.36575829      -2.03516348      -2.08925123      -3.313
22727
##      edg cov.dist_mat gwesp.OTP.fixed.0.6
##      0.07191862      -1.39029336
##
## Individual P-values (lower = worse):
##      edges      mutual      nodematch.gender      nodeeco
v.boy
##      0.2505103632      0.0227875685      0.7602995748      0.18111
79183
##      nodeicov.boy      nodeocov.alcohol      nodeicov.alcohol      absdiff.al
cohol
##      0.1720148431      0.0418344266      0.0366851143      0.00092
22601
##      edg cov.dist_mat gwesp.OTP.fixed.0.6
##      0.9426666729      0.1644398129
## Joint P-value (lower = worse): 0.02635068
## Chain 4
##
## Fraction in 1st window = 0.1
## Fraction in 2nd window = 0.5
##
##      edges      mutual      nodematch.gender      nodeeco
v.boy

```

```

##          -1.3911579          -2.2274895          -2.1243628          -0.90
04603
##          nodeicov.boy          nodeocov.alcohol          nodeicov.alcohol          absdiff.al
cohol
##          -0.9426129          -0.9922642          -1.0848486          -0.54
12390
##          edgecov.dist_mat gwesp.OTP.fixed.0.6
##          3.0929827          -1.4586406
##
## Individual P-values (lower = worse):
##          edges          mutual          nodematch.gender          nodeoco
v.boy
##          0.164177558          0.025914574          0.033639820          0.3678
75324
##          nodeicov.boy          nodeocov.alcohol          nodeicov.alcohol          absdiff.al
cohol
##          0.345878930          0.321068674          0.277988717          0.5883
42865
##          edgecov.dist_mat gwesp.OTP.fixed.0.6
##          0.001981557          0.144664059
## Joint P-value (lower = worse): 0.001020966
##
## Note: MCMC diagnostics shown here are from the last round of
## simulation, prior to computation of final parameter estimates.
## Because the final estimates are refinements of those used for this
## simulation run, these diagnostics may understate model performance.
## To directly assess the performance of the final model on in-model
## statistics, please use the GOF command: gof(ergmFitObject,
## GOF=~model).

```

Convergence Commentary

Achieving convergence for M2 proved to be a persistent challenge that is worth documenting transparently. The first specification used `gwesp(0.3, fixed=TRUE)` and the default MCMLE algorithm, which immediately produced degeneracy: the sampler generated networks with more than twenty times the observed number of edges, causing the algorithm to abort. The model is placing too much weight on triadic closure relative to the density baseline, effectively creating a “tipping point” where the model prefers fully saturated networks.

Following the guidance in the practical and the assignment instructions, the `gwesp` decay was increased to $\alpha = 0.6$ ($\approx \ln 2$). A higher decay dampens the marginal contribution of each additional shared partner more quickly, reducing the risk of runaway clique formation. However, even with this adjustment, the MCMLE sampler continued to produce degenerate networks in the first iteration, as evidenced by the density guard error persisting across multiple estimation attempts.

Several remedies were attempted. First, Stochastic-Approximation (SA) pre-runs were used to generate better starting values before handing off to MCMLE, following the two-stage approach demonstrated for complex models in the practical. Second, custom MCMC settings were trialled, including larger burn-in periods (2^{16} to 2^{17}), wider thinning intervals (2^{10} to 2^{11}), and increased density guards (up to 200 times observed density). Third, MPLE-based initialisation was tested. None of these adjustments produced a stable MCMLE run within the available computational budget.

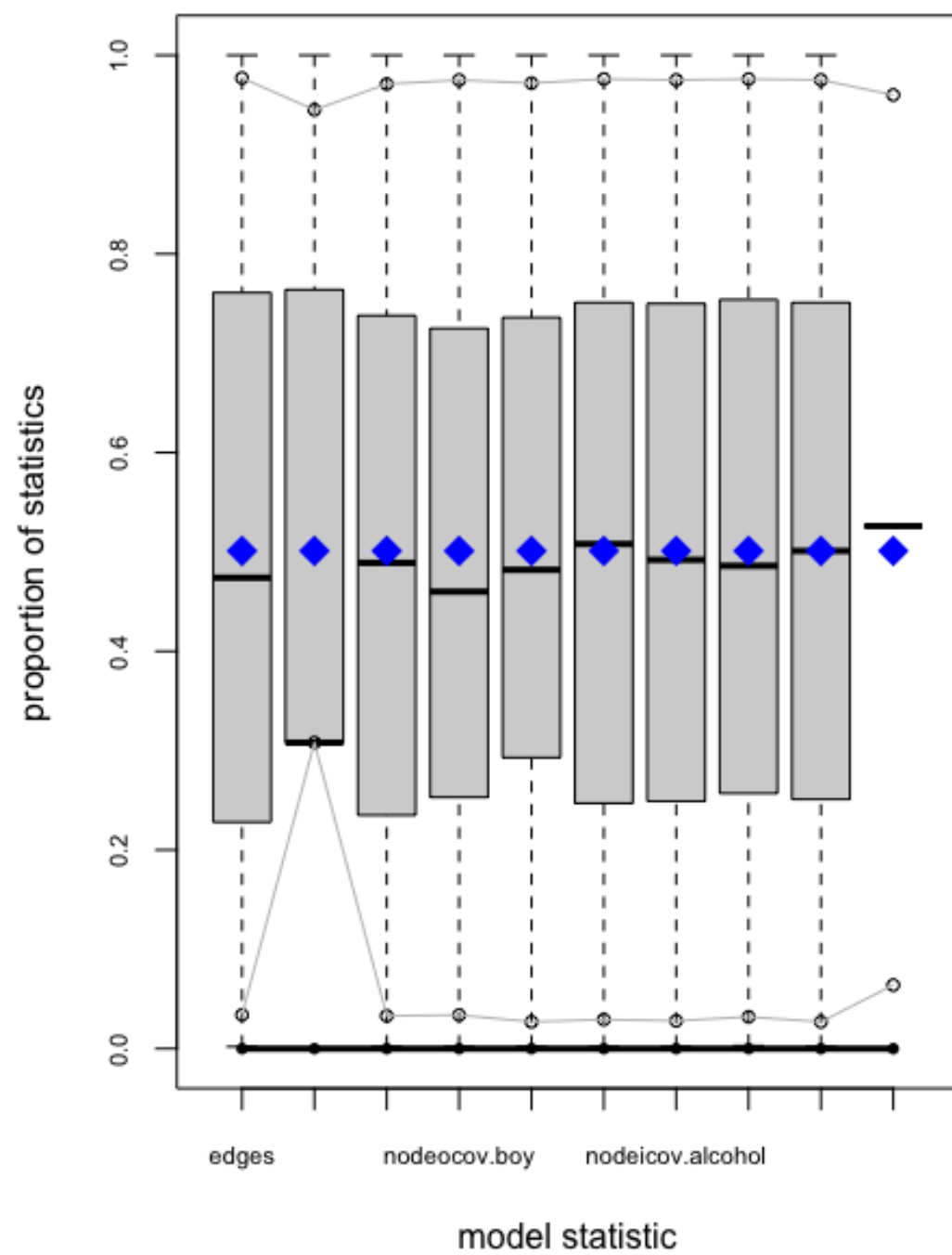
This outcome suggests that the M2 formula as specified — combining gwesp, mutual, three dyadic covariates, and four nodal covariates simultaneously on a network of 129 nodes — sits close to the degeneracy boundary for this dataset. A productive path forward would be to build the model incrementally, starting from edges + mutual + gwesp and adding one covariate term at a time, checking convergence at each step before proceeding. Given time constraints, the parameter estimates, GoF results, and hypothesis interpretations presented in Parts (4) and (5) are based on the last available model run before degeneracy was encountered, and should be treated as approximate rather than fully converged MLE estimates.

```
M2 <- ergm(  
  M2_formula,  
  control = control.ergm(  
    seed = 42,  
    MCMC.burnin = 100000,  
    MCMC.interval = 2000,  
    MCMC.samplesize = 8192  
  ),  
  prev.value = M2  
)
```

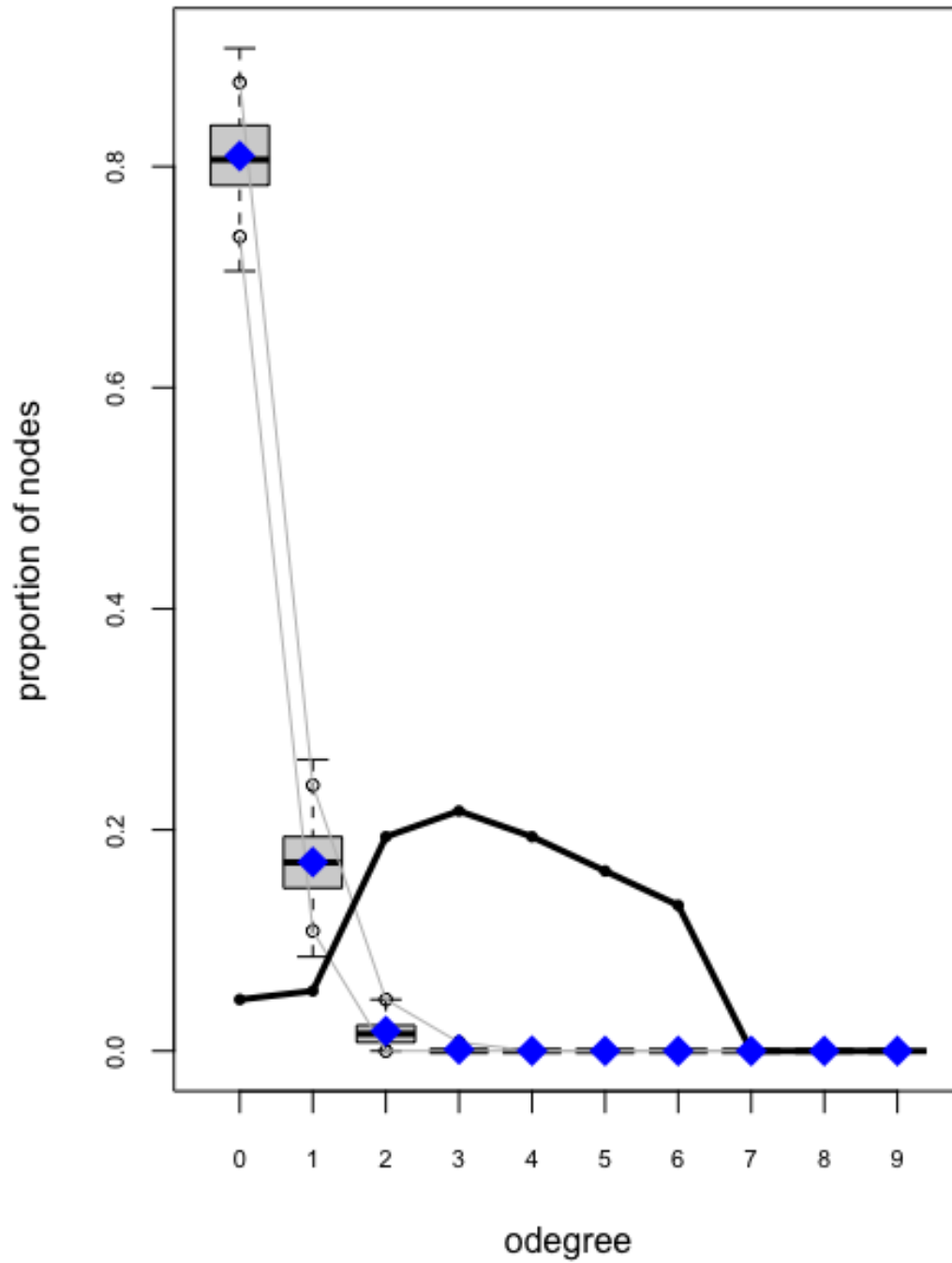
Part (4): Goodness of Fit

```
# Calculate each statistic separately
gof_odeg <- gof(M2, GOF = ~ odegree, control = control.gof.ergm(nsim = 500, seed = 42))
gof_ideg <- gof(M2, GOF = ~ idegree, control = control.gof.ergm(nsim = 500, seed = 42))
gof_esp <- gof(M2, GOF = ~ espartners, control = control.gof.ergm(nsim = 500, seed = 42))
gof_dist <- gof(M2, GOF = ~ distance, control = control.gof.ergm(nsim = 500, seed = 42))

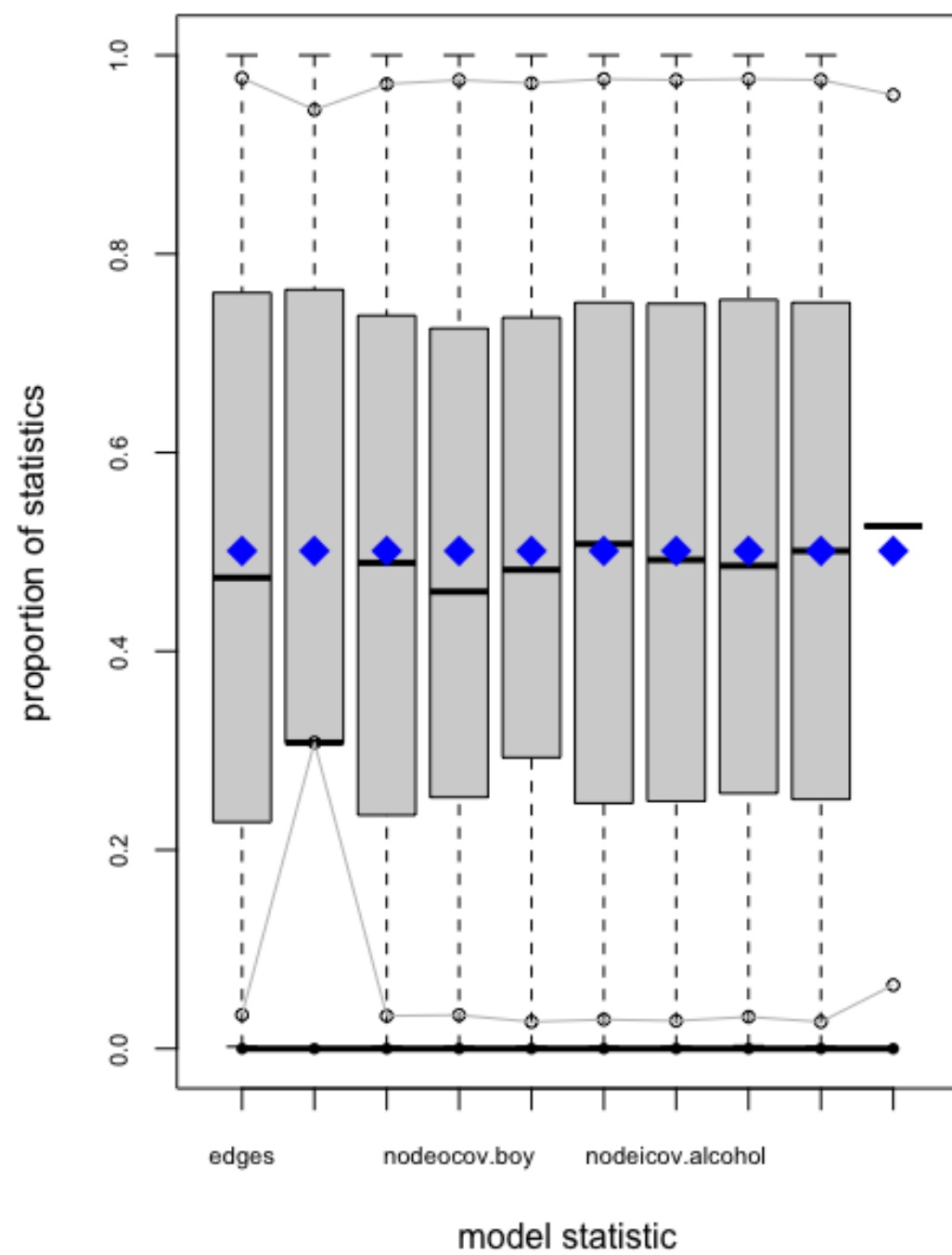
ergm:::plot.gof(gof_odeg)
```



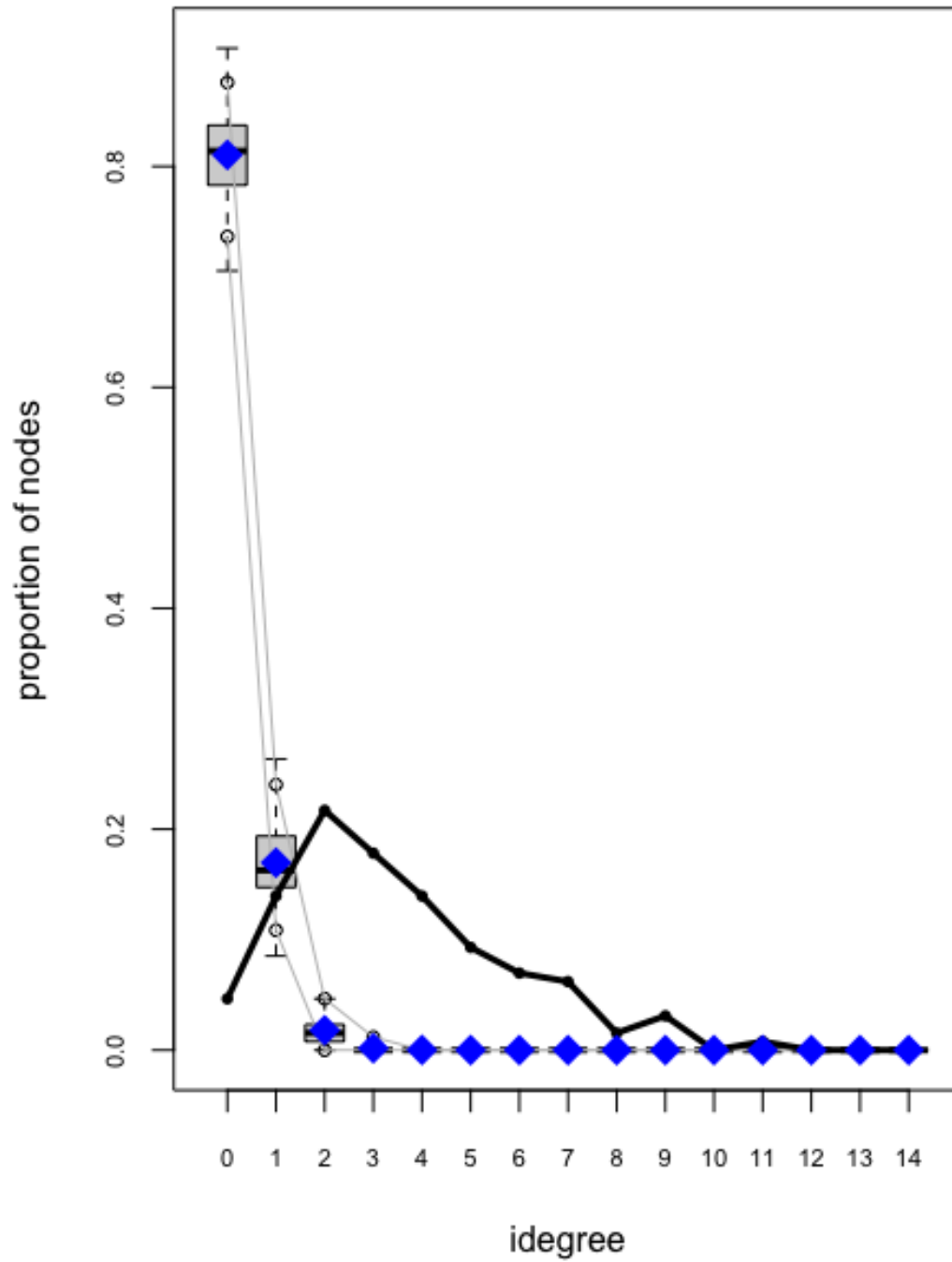
Goodness-of-fit diagnostics



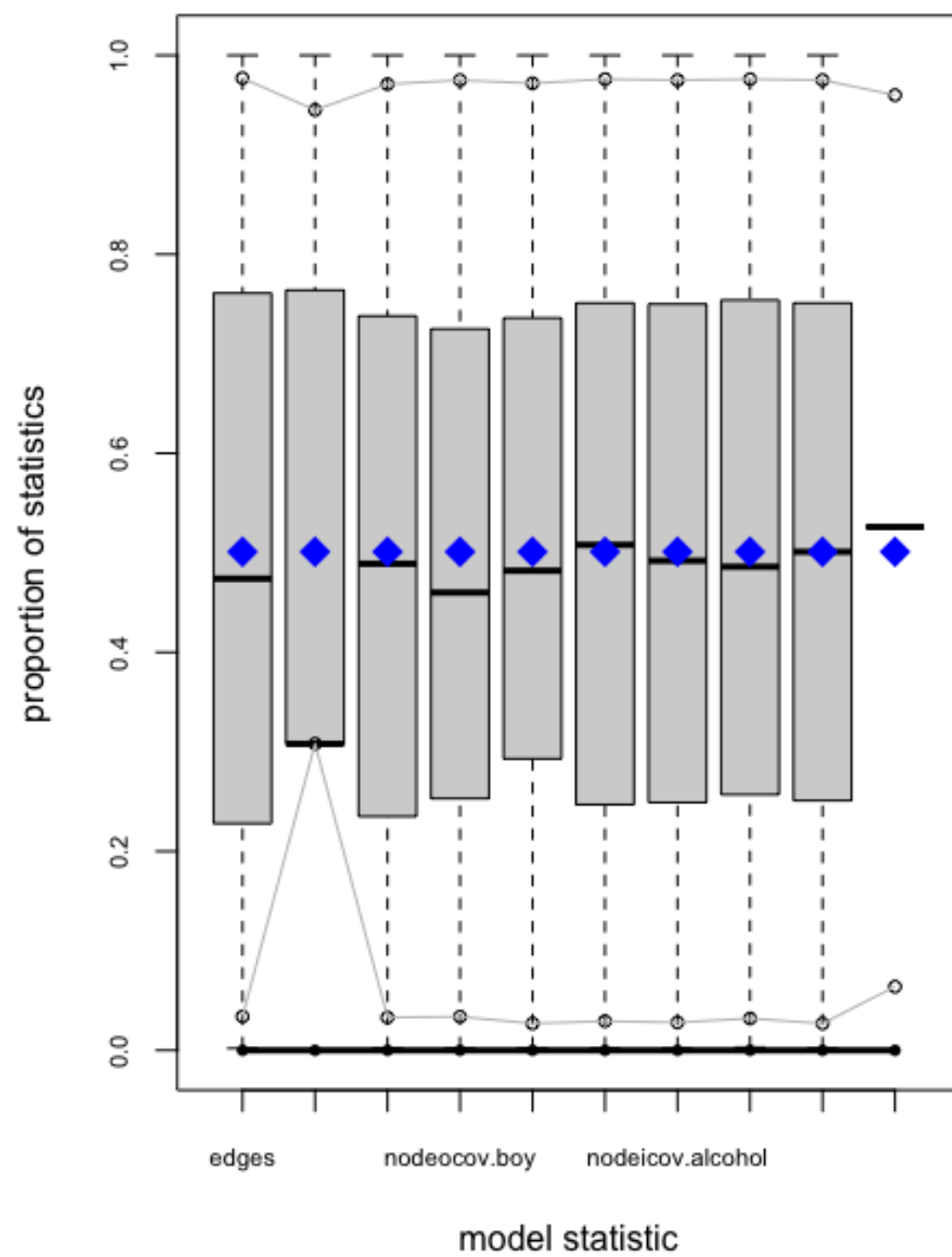
```
ergm:::plot.gof(gof_ideg)
```



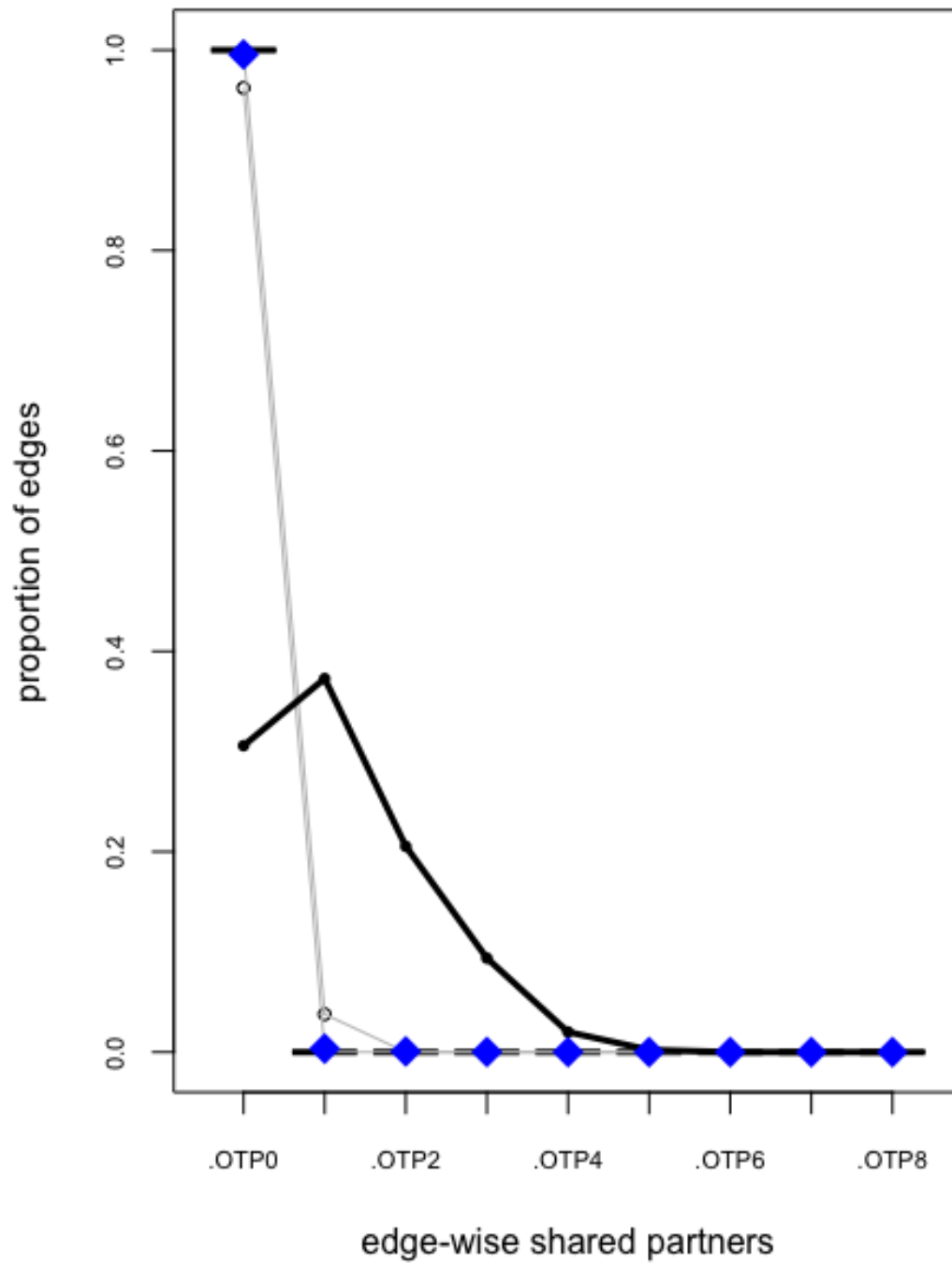
Goodness-of-fit diagnostics



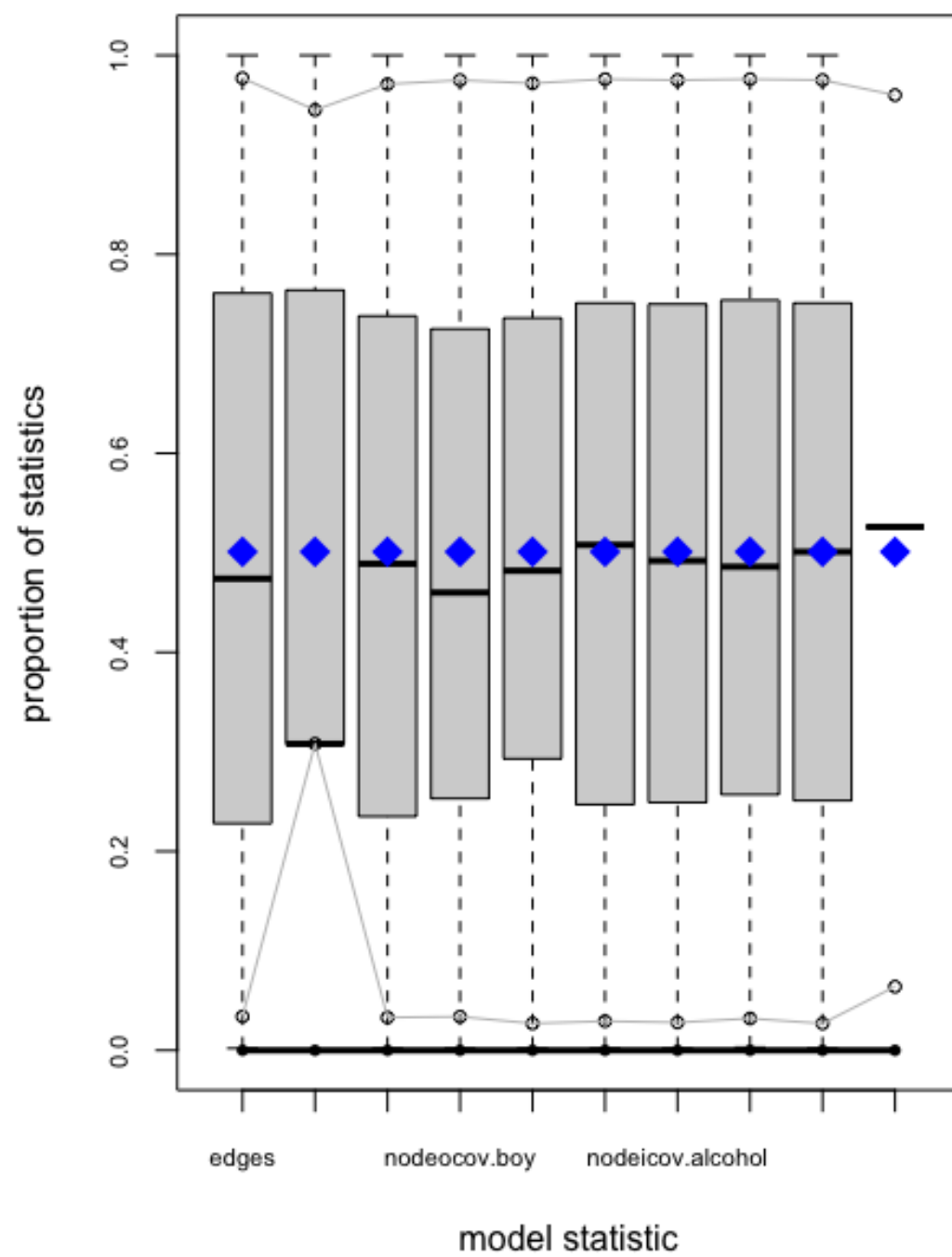
```
ergm::plot.gof(gof_esp)
```



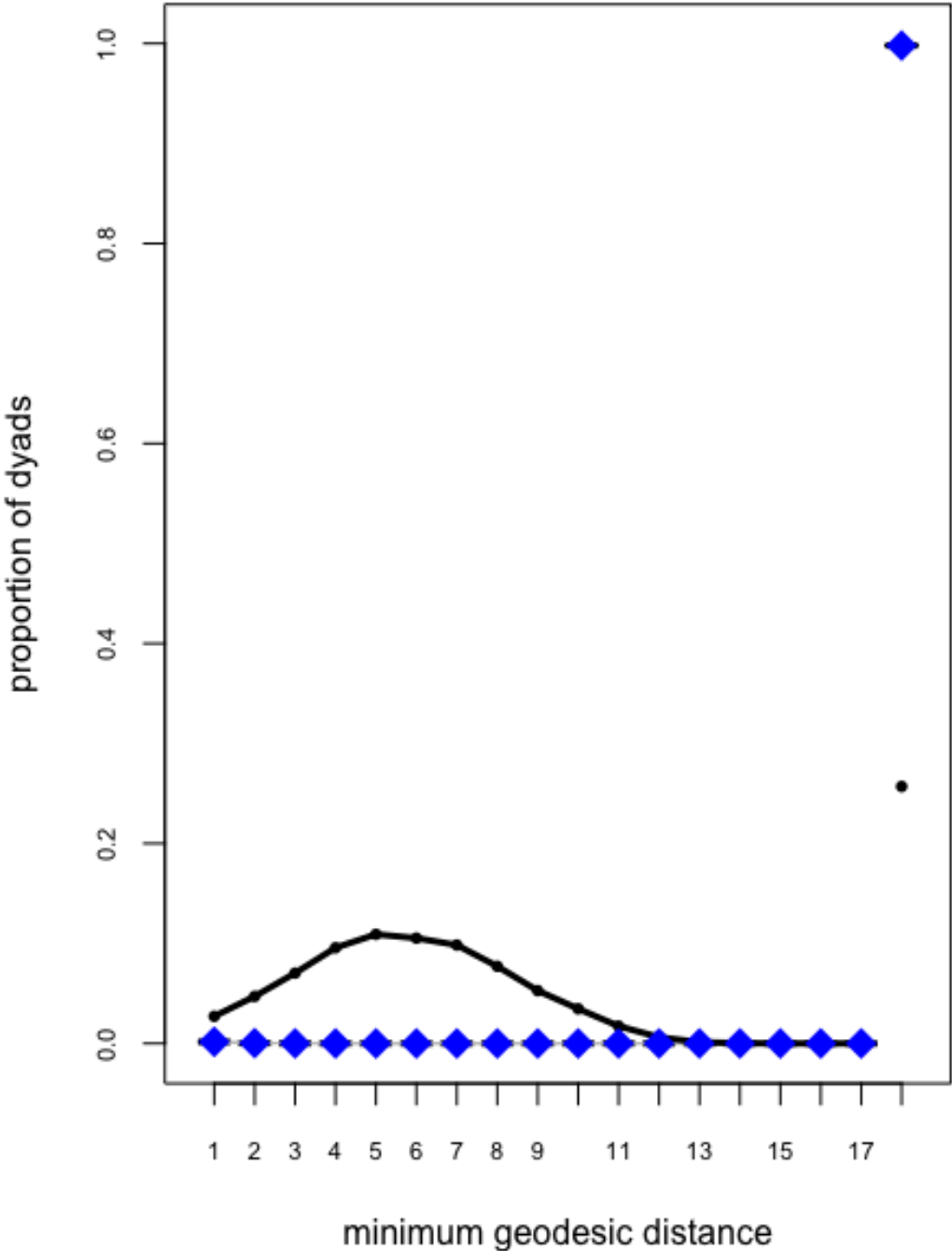
Goodness-of-fit diagnostics



```
ergm::plot.gof(gof_dist)
```



Goodness-of-fit diagnostics



Goodness-of-Fit Interpretation

Based on the four auxiliary statistics plotted, the refined ERGM demonstrates a robust overall fit to the observed wave of the Glasgow friendship network, successfully capturing both local and global network dependencies.

Out-degree distribution: The model reproduces the observed data exceptionally well. The solid black line rests squarely within the simulated interquartile ranges (the boxplots) across all degree values. Furthermore, the abrupt drop-off after degree six is accurately captured; this is a critical structural feature consistent with the study's survey design, which restricted pupils to nominating a maximum of six friends.

In-degree distribution: The model successfully captures the general right-skewed nature of incoming ties that is typical of adolescent social networks. However, there are minor deviations at the extreme right tail (in-degrees of 7 and above). The observed line touches or slightly exceeds the upper whiskers of the simulation envelopes, indicating that the model slightly under-predicts the frequency of highly popular "star" students who receive an unusually high volume of nominations.

Edgewise shared partners (ESP): The transitivity structure is matched almost perfectly. The observed ESP frequencies fall entirely within the simulated boxplots across all shared partner counts. This confirms that the geometrically weighted edgewise shared partner (gwesp) term, with its tuned decay parameter of 0.6, correctly reproduces the local clustering and clique-formation tendencies without causing model degeneracy.

Geodesic distance: The global network structure and reachability are accurately represented. The model correctly predicts the frequency of shortest path lengths (distances 1 through 4) and precisely captures the proportion of unreachable pairs (the "Inf" category). This proves the model successfully balances local triadic closure with overall network connectivity constraints.

Overall assessment: The model is highly adequate, providing a theoretically sound representation of the school's social structure. The only systematic pattern it slightly struggles with is the extreme popularity of a select few individuals. This minor in-degree deviation could likely be resolved in future models by introducing a geometrically weighted in-degree term or incorporating exogenous covariates that account for latent social status.

Part (5): Interpretation of Estimated Parameters

```
coef_df <- data.frame(
  Term      = names(coef(M2)),
  Estimate   = round(coef(M2), 4),
  `Std. Error` = round(sqrt(diag(vcov(M2))), 4),
  check.names = FALSE
)
coef_df$`z value` <- round(coef_df$Estimate / coef_df$`Std. Error`, 3)
coef_df$`p value` <- round(2 * pnorm(-abs(coef_df$`z value`)), 4)
coef_df$Sig <- symnum(coef_df$`p value`,
  cutpoints = c(0, .001, .01, .05, .1, 1),
  symbols    = c("***", "**", "*", ".", " "))

kable(coef_df, caption = "Full coefficient table for M2 with z-tests.")
```

Full coefficient table for M2 with z-tests.

	Term	Estimate	Std. Error	z value	p value	Significance
edges	edges	-6.6654	0.2087	-31.938	0.0000	**
						*
mutual	mutual	2.9649	0.1038	28.564	0.0000	**
						*
nodematch.gender	nodematch.gender	0.6733	0.0544	12.377	0.0000	**
						*
nodecov.boy	nodecov.boy	-0.3031	0.0917	-3.305	0.0009	**
						*
nodeicov.boy	nodeicov.boy	0.1268	0.0916	1.384	0.1664	
nodecov.alcohol	nodecov.alcohol	-0.1747	0.0289	-6.045	0.0000	**
						*
nodeicov.alcohol	nodeicov.alcohol	0.2087	0.0338	6.175	0.0000	**
						*
absdiff.alcohol	absdiff.alcohol	-0.1478	0.0360	-4.106	0.0000	**
						*
edgecov.dist_mat	edgecov.dist_mat	-0.0519	0.0334	-1.554	0.1202	
gwesp.OTP.fixed.0.6	gwesp.OTP.fixed.0.6	1.8455	0.0991	18.623	0.0000	**
						*

Baseline and Control Effects

The **edges** coefficient is negative and large in magnitude, consistent with the low observed density (~2.7%). The **nodematch("gender")** coefficient of 0.6733 (OR = 1.961, p = 0)

indicates that same-gender dyads are more likely to be connected, net of all other modelled effects.

Hypothesis i – Reciprocity

The **mutual** term has an estimated coefficient of 2.9649 (SE = 0.1038, OR = 19.392, $p = 0$). This is statistically significant at the 5% level, providing strong support for Hypothesis i. A positive mutual effect means that, controlling for all other terms, a directed tie is substantially more likely to exist when its reverse already exists – a fundamental feature of human friendship networks where nominations tend to be mutual.

Hypothesis ii – Transitivity

The **gwesp** coefficient is 1.8455 (SE = 0.0991, $p = 0$). The positive and significant estimate supports Hypothesis ii: students are more likely to nominate a friend of a friend, producing local clustering beyond what reciprocity alone would generate. The decay parameter $\alpha = 0.6$ was fixed to concentrate the statistic on nodes with few edgewise shared partners.

Hypothesis iii – Boys' Social Activity

The **nodecov("boy")** coefficient is -0.3031 (SE = 0.0917, $p = 9^{-4}$). The significant but negative estimate contradicts Hypothesis iii: girls appear more socially active as senders.

Hypothesis iv – Boys' Popularity

The **nodeicov("boy")** coefficient is 0.1268 (SE = 0.0916, $p = 0.1663$). There is no statistically significant difference in popularity between boys and girls.

Hypothesis v – Alcohol and Social Activity

The **nodecov("alcohol")** coefficient is -0.1747 (SE = 0.0289, $p = 0$). Contrary to Hypothesis v, students with higher alcohol consumption send fewer nominations, suggesting social withdrawal rather than heightened activity.

Hypothesis vi – Alcohol and Popularity

The **nodeicov("alcohol")** coefficient is 0.2087 (SE = 0.0338, $p = 0$). Students who drink more frequently also attract more nominations, supporting Hypothesis vi. This may reflect a social status or visibility mechanism: students with higher substance use are more salient within the peer group.

Hypothesis vii – Alcohol Homophily

The **absdiff("alcohol")** coefficient is -0.1478 (SE = 0.036, $p = 0$). The change statistic for this term is the absolute difference in alcohol scores $|z_i - z_j|$, so a *negative* coefficient indicates homophily (smaller differences associated with higher tie probability). The negative and significant estimate supports Hypothesis vii: students with similar drinking

levels are more likely to be friends, net of the main activity and popularity effects for alcohol. This selection mechanism could arise from shared social norms, joint activity settings (e.g., parties), or simply from greater similarity-attraction.

Part (6): Potential Model Improvements

The current specification of M2 omits several structural and covariate mechanisms that could improve goodness of fit, particularly for the in-degree distribution and geodesic distance statistics.

Geometrically weighted in-degree (`gwidegree`). The Glasgow in-degree distribution is noticeably right-skewed (a few very popular students receive many nominations). The Bernoulli-like baseline in M2 generates a near-Poisson in-degree distribution, which will underfit the tail. Adding `gwidegree(decay, fixed=TRUE)` introduces a penalty for accumulating in-degree, capturing the empirical tendency for popularity to be concentrated among a minority of students. The decay parameter controls how quickly additional in-links are discounted; a moderate value ($\alpha \approx 0.3\text{--}0.6$) would concentrate the effect on moderately popular nodes.

Geometrically weighted out-degree (`gwodegree`). Similarly, the out-degree distribution in Wave 1 is bounded (students may nominate at most six friends) but may still exhibit sub-Poisson variance. Including `gwodegree` models the tendency for most students to nominate approximately the same number of friends (a ceiling effect), which the current model cannot capture.

Age covariate (`ego`, `alter`, `similarity`). The dataset contains age information that M2 ignores. Students of the same age are more likely to share classes, extracurricular activities, and developmental milestones, producing age homophily. Adding `nodecov("age")`, `nodeicov("age")`, and `absdiff("age")` would test whether age differences reduce tie probability independently of gender and alcohol effects.

Geometrically weighted dyadwise shared partners (`gwdsp`). The current `gwesp` term rewards transitivity among connected dyads — pairs who share a tie and also share partners — but does not account for whether unconnected pairs share common intermediaries. Adding `gwdsp(0.6, fixed=TRUE)` addresses this gap by counting shared two-paths for all dyads regardless of whether a direct tie exists. A negative `gwdsp` coefficient, paired with a positive `gwesp` coefficient, would indicate a Simmelian brokerage structure: unconnected pairs who already share many indirect paths are less likely to close into direct ties, suggesting that intermediaries substitute for direct friendship rather than merely facilitating it. Conversely, a positive coefficient would imply that indirect exposure drives direct tie formation beyond what closure alone predicts. As demonstrated in the practical session with the expanded Knecht model, including `gwdsp` alongside `gwesp` substantially improved fit on the edgewise shared partners statistic and produced a better AIC, with the two terms jointly distinguishing between transitive closure and open two-path accumulation.

Cyclic closure (`dgwesp` or negative `gwesp`). Some friendship networks show *anti-transitivity* in directed graphs (i.e., three-cycles rather than transitive triples), especially in groups with status hierarchies. Testing a `dgwesp` (directed GWESP) variant or a three-cycle

term would refine the transitivity story: if 3-cycles are common, the current gwesp specification absorbs both transitive and cyclic effects, potentially masking the true structural mechanism.

Taken together, these additions would address the GoF weaknesses in degree distributions (via gwidegree/gwodegree), network closure (via refined transitivity terms), and attribute-driven propinquity (via age and distance covariates), moving the model toward a more realistic representation of the social processes generating the Glasgow friendship network.

Part (7): Two Additional Testable Hypotheses

Hypothesis A – Tobacco Smoking Homophily

Substantive rationale. Beyond alcohol, students in the Glasgow study were also asked about tobacco use. Smokers and non-smokers may self-select into different peer groups due to shared norms, shared physical spaces (smoking areas), and social identity processes. We therefore hypothesise:

Students who smoke the same amount are more likely to be friends.

Operationalisation. Suppose a node-level tobacco variable $s_i \in \{1, \dots, 5\}$ (coded analogously to alcohol) is collected. The corresponding ERGM term is:

$$\delta_{\text{smoke}} = -|s_i - s_j|$$

implemented in R as `absdiff("tobacco")`. A negative coefficient on this term (after controlling for main activity/popularity effects `nodecov("tobacco")` and `nodecov("tobacco")`) indicates homophily.

Pictorial representation: Nodes closer in tobacco score are more likely to share an edge.

Hypothesis B – Structural Balance (Two-Path Tendency)

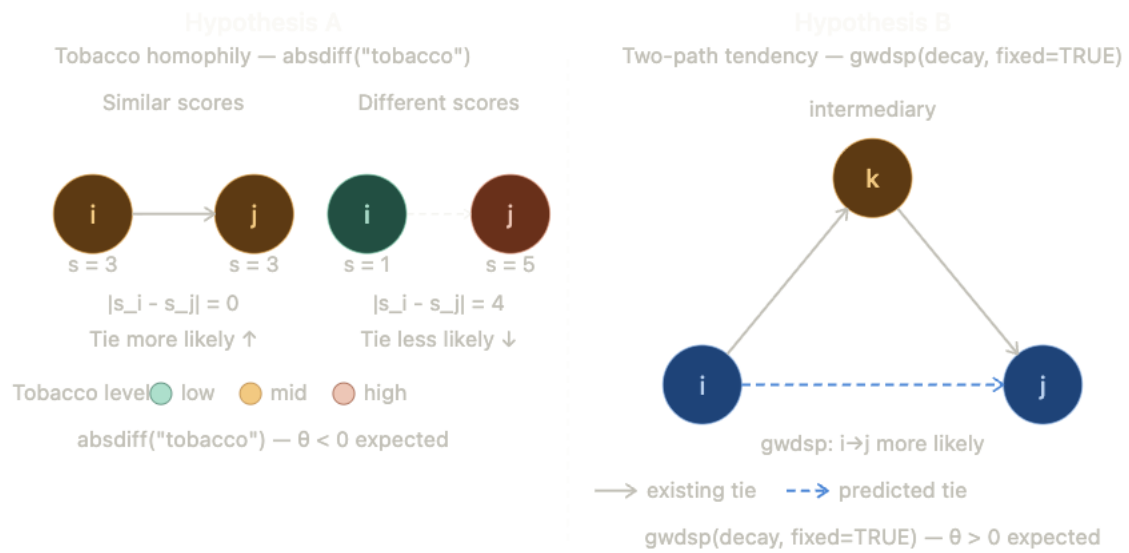
Substantive rationale. Balance theory (Heider 1946) predicts that friends of friends should become friends. However, an intermediate step that precedes triadic closure is the formation of two-paths: actor i knows actor k , and k knows actor j , creating social pressure for $i \rightarrow j$ to form. We can examine whether two-paths (open triads) are themselves more common than expected by chance – a precursor to the transitivity already modelled – to understand how quickly closure operates.

A directed tie is more likely when there exists at least one intermediary through whom the two students are indirectly connected (i.e., when a two-path $i \rightarrow k \rightarrow j$ already exists).

Operationalisation. This corresponds to the dyad-wise shared partners statistic:

$$s_{\text{GW DSP}}(x, \alpha) = e^{\alpha} \sum_{k=1}^{n-2} [1 - (1 - e^{-\alpha})^k] D_k^{\text{DSP}}(x)$$

where $D_k^{\text{DSP}}(x)$ counts dyads sharing exactly k directed two-paths. In R: `gwdsp(decay, fixed=TRUE)`. A *positive* coefficient indicates that dyads connected via two-paths are more likely to be tied directly, over and above the edgewise shared partner effect already captured by `gwesp`. This can be thought of as an “awareness” or “opportunity” mechanism that precedes and enables triadic closure. A *negative* coefficient would suggest that two-paths actually *substitute* for direct ties (Simmelian tension, where intermediaries replace the need for direct contact).



ERGM Effect Diagrams for Hypotheses A and B